Assignment 2:Problem Set – Signals and Systems

1. For the continuous-time system $y(t) = x(t)\cos(t)$, determine whether the system has memory, is linear, and causal.

2. For $y(t) = \sin(x(t))$, test linearity and memory.

3. For $y(t) = x(t-3) + x(2t)$, check causality and memory.

4. For $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$, state whether it is memoryless and causal.

5. For $y(t) = t^2 x(t)$, discuss memory, linearity, and causality.

6. For $y(t) = x(t+2)$, check if the system is invertible, time-invariant, and stable.

7. For $y(t) = e^t x(t)$, analyze invertibility and stability.

8. For $y(t) = x(t) - x(t-1)$, test time invariance and stability.

9. For $y(t) = \int_{0}^{t} x(\tau)\, d\tau$, test invertibility, stability, and time invariance.

10. For $y(t) = \dfrac{d}{dt}x(t)$, discuss invertibility, stability, and time invariance.

11. For the discrete-time system $y[n] = x[n] + 3$, check whether it is memoryless, linear, and causal.

12. For $y[n] = x[n-4]$, check memory and causality.

13. For $y[n] = x[n]\, x[n-1]$, test linearity and memory.

14. For $y[n] = \cos(\pi n)\, x[n]$, test causality and the memoryless property.

15. For $y[n] = \sum_{k=-\infty}^{n} x[k]$, test memory and causality.

16. For $y[n] = x[n+3]$, check invertibility and causality.

17. For $y[n] = \sum_{k=0}^{n} x[k]$, test time invariance and stability.

18. For $y[n] = x[-n]$, check invertibility and time invariance.

19. For $y[n] = \dfrac{1}{n+1}\, x[n]$, check stability and invertibility (assume $n \geq 0$ for input domain if needed).

20. For $y[n] = x[n] - x[n-1]$, test time invariance, stability, and invertibility.

21. Compute the convolution $y(t) = x(t) * h(t)$ for $x(t) = u(t)$, $h(t) = u(t)$.

22. For $x(t) = e^{-t}u(t)$, $h(t) = u(t)$, find $y(t) = x(t) * h(t)$.

23. For $x(t) = u(t-1)$, $h(t) = e^{-t}u(t)$, find $y(t) = x(t) * h(t)$.

24. For $x(t) = e^{-2t}u(t)$, $h(t) = e^{-t}u(t)$, determine if the resulting output is BIBO stable.

25. Verify the commutative property of convolution using $x(t) = u(t)$ and $h(t) = e^{-t}u(t)$ by computing both $x * h$ and $h * x$.

26. Compute $y[n] = x[n] * h[n]$ for $x[n] = u[n]$, $h[n] = u[n]$.

27. For $x[n] = (0.5)^n u[n]$, $h[n] = u[n]$, find $y[n] = x[n] * h[n]$.

28. Let $x[n] = \delta[n] + \delta[n-1]$ and $h[n] = \{1, 2, 3\}$ (i.e., $h[0] = 1, h[1] = 2, h[2] = 3$, zero elsewhere). Compute $y[n] = x[n] * h[n]$.

29. Verify commutativity of convolution with $x[n] = u[n]$ and $h[n] = (0.5)^n u[n]$ by evaluating both orders.

30. Show using the convolution sum that $x[n] * \delta[n] = x[n]$ for any absolutely summable $x[n]$.

31. Let

$$x(t) = u(t) - u(t-2), \quad h(t) = u(t) - u(t-3),$$

which are finite-duration rectangular pulses of length 2 and 3 respectively. Compute the convolution

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

and sketch the result.

32. Let

$$x(t) = \begin{cases} 1 - |t|, & |t| \le 1 \\ 0, & \text{otherwise} \end{cases}, \quad h(t) = \begin{cases} 1 - |t|, & |t| \le 1 \\ 0, & \text{otherwise} \end{cases}.$$

Compute the convolution

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

and sketch the result.